ActInf GuestStream 058.1 ~ Working with Gerald Edelman

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Hello and welcome everybody. It is September 19th, 2020-23. We're here at the Active Inference Institute in Guest Stream 58.1 and today the session will be around the experiences of those who have worked with Gerald Edelman. Edelman lived from 1929 to 2014 and as hopefully this conversation can explore and unpack a little bit, worked in various areas related to the brain, body, and mind. So without further ado, I will have our guests introduce themselves, give a little bit of a background on how they came to be working with Professor Edelman, and then we have several prepared questions and several other questions. So thank you all for joining and perhaps Andrew, if you'd like to begin. All right, so I'll just start a little bit with my background. I did my PhD at University of Minnesota and then I did my postdoc at Hopkins with Apostles George Opelis and Vernon Mott Castle and then I got a job in Phoenix working in a small neuroscience institute or neuroscience department at a hospital and I worked there for six or seven years and my interest was working with non-human primates with monkeys and studying how they reached for different objects and drew objects and then looked at activity in primary motor cortex and we carried out different kinds of analysis based on the directionality of the neurons in motor cortex, built populations, and were able to actually look at these neural signals and see how they corresponded in a pretty exact way to the way the animals are moved in time and space, in other words trajectories. And after working there six or seven years I was invited to come and look at this new institute that Gerald Edelman was establishing in San Diego in La Jolla and met a bunch of super interesting people like Giulio Tononi and Olaf Spornes and really their deep intellect and curiosity and creativity has convinced me to move to San Diego.

And so I ended up moving there and so Edelman was able to build this gorgeous research facility and he was able to build this project and the story on the Torrey Pines cliffs basically and it was a magnificent architecture

and he recruited a number of people besides Spornes and Tononi and so he divided the group into theorists and experimentalists and I was an experimentalist

and I was really exposed to a whole new way of thinking about neuroscience in terms of sort of the cognitive aspects of it,

which I still have difficulty with to tell you personally because I don't even know how you define cognition

and I'm not sure that anybody really can or as Vernon Moncastle would say, I don't know the answer to that nor does anybody else.

Maybe Carl would take issue with that.

But anyway, the coolest thing about it was being really exposed to a lot of creative thinking and deep intellectual endeavors

and I've really never regretted the seven or eight years I spent there.

I think it really helped change the way my career went from being sort of a motor control, strict, sort of, you know, very careful scientist to using my imagination a lot more than I would have otherwise.

So I'll leave it at that.

Awesome.

Cheng Ming?

Okay, yeah, I can speak about my journey with Dr. Ederman.

So I was originally from Taiwan.

I graduated with a medical degree in Taiwan University in 1978, but I was very interested in doing basic science.

So I applied the PhD program and got into Rockefeller University and that started to search for a laboratory.

And certainly Dr. Ederman, you know, with his very powerful intellectual capacity building

in the building, you know, it was on ninth floor of the Bronx laboratory.

I was very impressed. So I joined the laboratory.

So since 78 to 83, I was a PhD student.

And I was working on the topic called called neural cell adhesion molecule.

So at the time Ederman already got his Nobel Prize for immunoglobulin.

But he would be, you know, usually, you know, in usually elegant way would say, I solve the molecular recognition.

So he said, I don't want to be staying in immunology to collect more awards.

He said, I will work out the cell recognition.

So in his mind, the cell, how cell recognition work is through a group of adhesion molecule interaction.

So that's how he started to work on the cell adhesion molecule.

And the Dutis Hauser actually worked out the chicken cell adhesion molecule.

So when I joined the laboratory, my PhD things were, you know, work out the first mammalian cell adhesion molecule.

And in addition to N-Chem, though, I supposed to work out on these adhesion molecule mediating the neural cell recognition.

As he said, he would say, you know, I asked him, oh, so after all, what would be my PhD thesis?

And he would say, Ming, you know, when I start to write a poem, I don't know how it's going to end.

You know, that's the way he would put his language there.

So he said, just figure out retinal tactile projection, you know, so that, you know, I'm a hardworking student.

So I would work out on that.

And as you know, these days, our so-called systems biology approach, right, is these transcriptome single cell RNA sequencing.

To identify a non-biased way to identify molecules involved.

At that time, the so-called systems biology approach was making monochromal antibodies.

So I would be there taking out the brain, you know, and the retina and the tecton and produce,

I produce 18 batch of the monochromal antibodies against these neuronal cell membranes.

So I certainly have identified a lot of adhesion molecules.

But none of them really show a very distinct pattern.

I mean, these days, I think we, I mean, the field using the transcription factors was able to identify

some patterns, but at the last day, though, it's kind of all, all together.
But I remember that actually, no, so anyway, that's enough.
I didn't really figure out retinal tactile projection, but that's enough for me to graduate.
But at the last time, though, I also still remember, you know, how I actually, I mean,
still, it has a great impact on my professional career.
So I was cutting chicken spinal cord, you know, cross section to mapping these cell adhesion molecules.
So I have some feathers, chicken feathers hanging outside there.
And they stand with this end cam with the most beautiful, exquisite pattern.
You know, I mean, brain, I later work on the feather as my major research model,
kind of as a Rosetta song to understand the language of morphogenesis.
But there, I remember how I started, it said you have these feathers, they form these branches.
So these branches are actually originally formed cells, then they die to become space.
But before they die, they actually light up with the end cam, light up with the end cam,
with the most exquisite pattern there.
So you kind of say, okay, you're going to carve out this region.
You light up with end cam, you light up with the sonic, and then other signals come in to say you die.
But before they die, they light up that pattern.
So the feather pattern is very exquisite, radial symmetry, bilateral symmetry, bilateral asymmetry.
So Edelman was very impressed by those, you know, those findings.
And so that, you know, that I appreciate that he actually encouraged that.
He, he, when he sees something he appreciate, he, he, he go, he, he encouraged it.
He didn't say, oh, you supposed to work on the brain or whatever, he actually like it.
And subsequently, he also, you know, developed this idea of topobiology, which he would acknowledge to say,
it's sort of a based on the very exquisite pattern we found in the, in the feather.
But of course, he developed it further.
But some other time he would say, if this is not in feather, I'm going to, I am Edelman,
I'm going to get another Nobel Prize.
So I, I like it that he encouraged that direction.
So later, when I became independent, I sort of say, okay, you know, um, brain is wonderful,
but it's all like a being cut inside.
I don't figure it out, but I'm not that smart, but I might be able to figure out the feather pattern
formation.
So I have a reasonable, successful career, uh, figure out, you know, the able, debo of the feather.
Uh, but, but as I said, not just the feather itself, but use feather as a, as a, as a Rosetta
stone to figure out morphogenesis.
And, uh, uh, that, uh, sort of, I think, uh, I appreciate that, uh, his encouragement on that.
And, uh, also, you know, as you know, the same cell biology comes along and the feather in a bird,

you know, baby chicken is one way about chicken, male, female, all looks differently, but they are all from the same group of the stem cell that my lab has demonstrated.
So it is the so-called stem cell niche would be modulated by sex hormone, by age, by season, by temperature, and then they help you produce different, uh, feather forms to, to let, to adapt to the environment.
So, so I, I think I, I enjoy my research career and it was obviously, I have his impact and I think he's intellectually, uh, brilliant and always encouraged going after, after fundamental principles saying that if you're going to work on a question, ask the most profound questions and approach with the most available technology, I mean, the technology of the day, you know, don't waste your time on, on small questions.
And, and I think I have to take those in.
So I can stay up to here and we can discuss more later.
Thank you.
Awesome.
Carl.
Hi.
So my name is Carl Friston.
Um, like my two colleagues, I found my stay with Jerry Edelman, both foundational and very, very formative.
Um, I was quite early in my research career when I joined the Neuroscience Institute.
Um, having trained as a psychiatrist, I got into brain imaging and at the inception of brain imaging, people were for the first time discovering principles of functional brain organization, such as the principle of functional segregation and subsequently principles of integration, how segregated, um, specialized parts of the brain communicate or are coordinated or are integrated together.
And one of the key theoreticians and neuroanatomists that, um, authored this particular interpretation and deployment of neuroimaging was Semi Zeki.
And Semi Zeki, uh, those of you who don't know, he's, he's still very active.
Uh, uh, one of the architects of the, of neuroaesthetics, um, um, in his late retirement now, uh, but he was friendly with Jerry Edelman and Semi was, um, famed for, um, understanding the computational anatomy of early visual cortex and visual cortical hierarchies and the, um, and I repeat the principles of functional specialization and integration.
Um, so after a couple of years, um, at the inception of human brain mapping using positron emission tomography and establishing, um, this, um, for example, the color center in the, in a particular part of the visual cortex, I was effectively sent to New York.
Um, so my period of, um, engagement was a, um, a fellowship between about 1990 and 1992, 93.
So this slightly predates Andy, um, and it covered the, um, translocation of the neurosciences Institute from the Rockefeller university to, um, uh, La Jolla, um, and, uh, the, uh,

under the auspices of the script.
So I spent, you know, half, uh, my fellowship at the Rockefeller and then subsequently, uh, watched the building of the magnificent neuroscientist Institute on North Torrey Pines Road, that Andy referred to with the auditorium, if I remember, designed by imported, um, experts in audition and music and sound engineering from Sweden.
It was a really an amazing place.
I, I do remember a couple of lunches, um, before I returned to the UK to, uh, pursue my career in imaging neuroscience, um, under the auspices of the Wellcome Trust.
Um, but only a couple, because we were sort of, um, in rented accommodation from the scripts on the other side of the road, uh, but it was an awe inspiring, uh, bit of architecture, both intellectually and, um, practically in terms of, uh, the, you know, the early days of the Neuroscience Institute.
Um, so I was sent there basically, uh, because, um, uh, Sam Izaki thought it would be good for me to go and work with Gerry Adenman.
Um, retrospectively, I now appreciate that I was, um, sent there to replace a gentleman called Reed Montagu, um, who had been tasked with on the pure theory side, as opposed to the cell adhesion molecule and the, the wet lab, um, um, side of, uh, Edelman's research, um, on the pure theory side, um, uh, the point of my arrival and to a certain extent at the point of Reed's, um, brief, uh, exposure, um, to the Neuroscience Institute, the theory of neural group selection was becoming consolidated, and in particular, what drove the selection of neuronal groups, neural ensembles, cell assemblies, however you want to describe them, um, and given, um, um, Edelman's, um, penchant for incredibly incisive in, um, and, um, straightforward, but very creative biological thinking, he thought, well, it's obviously some kind of natural selection, um, and therefore there must be the homologue of adaptive fitness that is underwriting the scaffolding and the dynamic, um, uh, structuring of connectivity or connections that, um, were definitive definitive of in the spirit of functional specialization segregation, but also the mediator of functional integration, um, of these, uh, neural cell, uh, neural groups, so this, um, this became known generically as value, um, so Reed was working on value-dependent learning, the plasticity and the principles that underwrite which synaptic connections have the right kind of adaptive fitness in their milieu, in that context, that enabled them to persist and other, uh, connections went away and became, um, um, or contributed to the, you know, the sparse coupling of, of neural networks, um, and then Reed, um, and we can talk about the, you know, the glorious stories about the, the, the departure of Reed, um, Reed actually ended up, um, um, in a, in a rather circuitous way with, uh, Terry Sonosky at the Salk, um, subsequently then hooked up with people like Peter Dian and Wolfram, um, Schultz, uh, who at that time was in Zurich, and of course this, the, you know, the story that emerged from that was the reward prediction error story, um, that, um, actually ironically, uh, Reed did not, but a number of, uh, Wolfram, um, eventually got the brain prize for some decade or so later, but the inception of that story was actually at the Rockefeller, um, and it was, um, basically, um, a, an attempt by, I think Reed to formalize the, the principles are, that must be in play to make the theory of neural group, um, selection under what were important, what was an important,

um, dynamic from, uh, Edelman's point of view, which is the notion of re-entrant dynamics, so very, very early on, um, he had, um, identified the, the fundamental importance of, of recurrent connectivity and re-entry as very formative in, um, the, you know, the, the emergence of these structured groups and the, um, associated functional specialization, um, and I'll just open brackets just to point out that, you know, you could, um, you could claim that certainly his early conversations with, with, um, um, people at that time, sort of preempted stuff in computational neuroscience that we now would take as, of, uh, for granted, not really realizing where it came from, but certainly his work with Vernon Mancastle, I'm sure Andy will be able to speak to this, uh, later on, on, you know, the importance of re-entry and dynamics and oscillations in the formation of these groups and sort of cortical columns and the like that was a sort of, I think, um, you know, the, the monographs that they, not the monographs, but the, the, the, the papers and, uh, the articles they wrote at that time, I think were quite visionary. Um, Steven Grossberg was also hanging around a few years earlier, and I think you could probably claim that between the two of them, they invented computational neuroscience, certainly a la, you know, New York, Boston, sort of, uh, you know, the East Coast kind of, um, um, style of, uh, of computational neuroscience, mostly involved with something called the Neuroscience Research Foundation,
so that they, as I understand it, and again, Andy may want to correct me here, but this was like the, um, the forerunner of the Society for Neuroscience, it was a group of people, people, um, and perhaps Max Cowan is, is the name that comes to mind, people who actually invented words like neuroscience, uh, which wasn't a thing, certainly when I was applying to, to, you know, to, to my undergraduate studies, um, and trying to create institutions, academic institutions that would endorse this notion of neuroscience, and the Neuroscience Research Foundation was like the old guard, and then the young Turks came along and reacted against the, um, the old guard, the Neuroscience Research Foundation to form the Society for Neuroscience, and then we all know the story, uh, since that time, it's now one of the biggest societies in, in academia, um, so it's just making the point that Edelman and, um, you know, his colleagues and vicarious colleagues, uh, you know, a lot of people, Oliver Sacks being, being one, Ernst Mayer being another of his favourites, sort of, um, um, characters to talk to, all of whom would come and visit, and we were privileged just to listen in and conversations in this very formative time where some of the, sort of, I think, conceptual pillars of the way that we understand, you know, neuroscience, uh, um, uh, you know, in the 21st century were, um, you know, were first, um, articulated, probably not using quite the same words that, that we use nowadays, so no one talks about re-entry anymore, uh, sort of about recurrent neural networks, or sort of, um, uh, feed forward and feed back and the like, uh, but the same underlying dynamics and maths, I think, is absolutely there. Anyway, I was brought in to replace Reed Montague, um, so that was my little contribution, um, whilst we're in the Rockefeller, um,
I made the mistake of, um, thinking that the right way to do this was to actually write down equations and try and sort of naturalize it in terms of, in terms of mathematics. Edelman was a great biological thinker, he was not a very good mathematical thinker, um, so there was a bit of a clash there, so

I, I remember one, uh, one meeting, um, very early on when it was being decided what I was going to be doing, um, and perhaps it's useful just to say the context in, you know, in which those, those early meetings took place. So, Andy has already mentioned Giulio Tononi and Olaf Spawns, they were two key, uh, people, uh, Edelman's young men who already, who already in, in situ, um, and then I replaced Reed, um, and they were working really purely on theory, um, and that theory was tested in the context of, uh, what was, uh, an, uh, forerunner of supercomputing, a hypercube that was driving, in my day, little robots. So again, you know, Edelman was doing neuro-robotics in the 19, you know, 1980s, um, in exactly the same way that people now in developmental neuro-neuro-robotics are, are, are still addressing exactly the same questions, um, probably with, with, um, the same kind of, uh, computer infrastructure that obviously upgraded. Um, so, um, we were sitting around and Edelman was, uh, called out, take a, uh, a phone call from, from Germany. Um, and there was a look of horror on people's faces because I'd suggested a relationship between value-dependent learning and the Rischkauer-Wagner rule, uh, uh, from behavioral psychology. Um, and as soon as I, you mentioned an equation or a formula that set Edelman off and I didn't know that. So I didn't know I pushed the wrong button. So when Edelman came back in, I was effectively rusticated. I was prevented from doing any work, um, for three months and for every day, afternoon and morning, I had to sit in the Rockefeller library and read purely biological great texts. Um, and I ended up having to, the last one was the growth of biological thought by Ernst Mayer. I had to read from, by, from page to page. And when I did so, I had become sufficiently sanitized to be allowed back into the workplace to actually do some work. Um, so that, that, that was our contribution. Uh, and then when we moved to, um, you know, the whole, um, parade of 16-wheeler trucks is translocating all the kit and all, all, all the paperwork to, um, to La Jolla and then relocating out there, settling down. Um, I brought that work to closure.
And the interesting, that was at the point when Julio, well, Olaf, I remember in saying, you know, what he wants is basically, um, to understand the, and simulate, um, the, uh, the brain, um, in a way that, uh, is, um, draws explanatory power from the architecture of the connectivity.
It basically would say he wants to be, he wants to establish connectomics, um, uh, as, as a discipline.
And of course, you know, 10 years later, he's Mr. Connectomics. Uh, Julia at that point was, um, inventing integrated, um, integrated information theory, um, with a particular focus on, um, information theoretic formulations of complexity and what special, if you like, counterintuitive behaviors do you get from these biologically configured connective when you put dynamics on structure and trying to quantify that in a way that ultimately, um, transpired to be measures like phi in IIT. Um, so that, that was a fun time, uh, as the mathematician, the closet mathematician, because I wasn't allowed to, to, to say anything about the maths at the time. Uh, I enjoyed that bit.
And so looking, um, you know, um, being in, uh, at the, if you like, um, ground zero at the inception of all these, all these aspirations, which eventually led to very fruitful careers. And then I went back, um, to, um, uh, to the UK just had fond memories and, uh, non-fond memories, uh, but both of a nostalgic sort since that time.
Awesome. Great tales. Changminger, Andrew, anything you want to add, or I can go to one of the questions.

Well, uh, just to, uh, expound a little bit on what Carl was saying. I don't know if you can
see this book. Okay. So this was, uh, something that came out of the neuroscience research program.
That was sort of the, uh, old boys club that Francis Schmidt, um, started, uh, at MIT and later
was the basis for the Neurosciences Institute. And what this was was sort of the who's who in
neuroscience. So again, if you look at this book, it says edited by Francis O. Schmidt. So he was the
guy who started that. And, uh, so this persisted, uh, even to the, uh, throughout the two thousands.
Um, and, uh, so once or twice a year, uh, at the Neurosciences Institute, they would organize one of
these meetings and it would be a who's who of neuroscience. So all the Nobel laureates in
neuroscience, all the big names would come for a four or five day meeting, uh, and talk about their
research. And as a member of the Neurosciences Institute, I got to sit in and meet these people.
And it was, uh, phenomenal, especially for, with my background, because I was sort of, uh,
not on the, the mainstream of neuroscience at that point. And, uh, that was, uh, really fascinating.
So you got to meet people like Will Schultzen and all the big names, um, that were there, Francis Crick,
um, you know, and, uh, but what I want to go back to is this original book by Edelman and Mockcastle
called the mindful brain. This is sort of like one of the, uh, Bibles of early creative thinking
about brain function. And it resaged the idea that of distributed systems. This was sort of the
thing that Vernon Mockcastle was very much into. Um, and so the, to this day, the way most people think
about brain function is, uh, boxes, uh, where each part of the brain is confined to a box.
And then there are arrows pointed to other boxes. And so we get this idea that there's a, a circuitry.
So you'll often hear neuroscientists talk about circuits and that kind of comes from the idea that
you have these boxes and you have arrows and you have plus and minus signs with the arrows. And then
somehow the pluses and minuses add up and tell you what's going to happen at the next box. Um,
and that is, um, nice because you can kind of simplify things and talk about causality
in, in very concrete terms. And it's great if you're an engineer, because you can write
transfer functions about defining how you can output from an input, but that's not the way the brain works.
And, you know, I, I think Kyle can attest to the early days of, uh, MRI, uh, studies where everybody
was doing pseudo colored, um, uh, representations of these different areas to see which area would light
up in particular task as if each area had a specific job and could be assigned to a specific task. And
Dr.
So neuroscience in general is still suffering from that point of view.
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now is finally starting to evolve into something that's becoming more mainstream.
And if you sort of look at the evolution of neuroscience, I think if you project ahead to the next 10 or 20 years, this is going to be really the fundamental change in the way we think about neural function.
And I'm super excited about that.
And you look and sort of back in history to see how that's evolved.
And I don't know.
I think there's something very attractive about thinking about things that way.
Yeah.
Okay.
Yeah.
Okay.
I'll add something.
I think talking about the, you know, Edelman's so-called, the brilliance about, you know, seeing things through.
I don't know in the neurobiology field that you people still sort of seeing the neurodarwinism idea is still there, right?
Yeah.
So that was the time when I was with him in Rockefeller University.
He would talk things like that, you know, Darwinism for evolution species.
And he saw that in immunoglobulin coronal selection.
So he applied to the, to this neurodarwinism idea.
But I have to say, you know, subsequently, I actually, my career is mainly in University Southern California and working on this morphogenesis pattern formation, tissue pattern formation.
And in our field, you know, sort of a more bench work, you know, aspect of Edelman, you know, this direction, not the neurobiology direction.
But in the field, there were people originally more talking about the stem cells are predetermined and then they go this lineage, they go this lineage.
But we using the skin as a model, actually, you know, I start to write these things, kind

of a Darwinism within the skin.

I mean, basically, it's that there is no preset forever stem cells.

If you wipe out these stem cells, other cells are waiting there to take over the space.

So there is even during development and during regeneration to make the tissue pattern process robust.

There is a small so-called Darwinism competition going on in there.

And that is sort of also influencing our work in morphogenesis these days.

And I think that's kind of the impact of him on this aspect as well.

Yeah.

Wow.

All right.

I'll read a question.

This was from Dave Douglas.

Dave wrote,

We know that Professor Edelman used deep analogies in exploring new subjects as in his leveraging his knowledge of immunology and organismal development to probe conceptualization and learning.

Did he ever work specifically on metaphor and analogy as such?

For example, did he leverage his experience of music and poetry to dig out scientific facts while separating fertile from sterile analogies?

An answer or just what that makes you think of?

It makes me think of something which Edelman used to call me, which I am an intellectual thug.

So I'm sure that there is a great answer to that excellent question.

And it does speak to Edelman's skills and interests as a violinist.

I do remember BBC Horizon came to do a profile on him and his ideas.

And it was prefaced with, I think, shots of him playing the violin, you know, to concert standard.

But I have absolutely no idea whether he used that sort of part of his life to take things beyond metaphors.

So as an intellectual thug, I'm going to hand over to my two colleagues to see whether they can give a more elegant answer.

Or anything in that area of the arts and you brought up poetry.

Anything in the arts and sciences?

Well, I'm not a brain biologist.

I just say that from my side of the interaction, you know, I mean, he really tried to merge science and art.

He would think, you know, a real science would be beautiful.

And that's the way I think when I learned, as I mentioned, you know, when I say, oh, I was expecting a more clear cut thesis idea.

But he would say, you know, Ming, it's like writing a poetry.

He said that when I start, I don't know where it's going to end.

You know, it's like some artists.

When you ask them, how do you create this painting?

I try to ask this question to some of my friends who are artists.

And they say they say they do not.

They do not have a clear blueprint of how that painting would look like.

They have an idea.

It's kind of floating in the sky and they try to grab it down.

So they paint on the way and try to make it into a painting.

Or I have a friend who are poets.

I ask them this.

They say they grab the word to make them coming down.

And I kind of feel Adam, I sort of treat the scientific idea, idea like that.

You know, this is my superficial, you know, interaction.

Yeah.

Yeah, just some experience.

Well, I don't have a correct reply to that.

But, you know, in general, I think there's the idea of creativity that he brought to the forefront.

And then also, you know, with that creativity, you have to have a certain amount of courage to move out of the sort of mainstream, right?

So if you have an idea and you have to have a certain amount of courage to put that forward, that idea is creative and out of the ordinary.

And I think artists and sort of really great scientists have that in common.

Maybe to that nexus of creativity and career development and courage, for you or just in the environment, how did his mentoring help bring out these threads in people?

For better and for worse.

I think Carl could attest to that.

I have a hard time viewing Edelman as a as a feel good mentor.

I think he would throw ideas out there.

He could be pretty aggressive and.

I would say frightening to young people.

So I wouldn't.

I wouldn't cast elements behavior as sort of the typical mentor.

But again, I think the environment that he created was someplace that was really special and reflected that kind of courageous creativity.

And I really appreciate that now.

Just to endorse that notion of courageous creativity.

I think it's a really nice observation.

I've often reflected on whether.

Let's take, for example, Julio and Olaf.

And I think you could probably apply to me and Andy and a number of other people as well.

But if we just take Julio and Olaf, who both become world leaders, thought leaders in their own particular field, having brought something new to the table.

And in some in some instances in more than one area.

So Julio Tannone in terms of sleep research and synaptic homeostasis, in terms of his work in consciousness studies, integrated information theory.

Olaf in terms of structural function and connectomics.

I mean, is it that Edelman had a good eye for bright young men or is it that bright young men learned how to do this, that this was the way to be courageous and to set an adversarial in that courage to be creative and to pursue what you have created in a slightly adversarial or at least sort of defensive way?

I don't know.

I don't know.

But what I do know is there was certainly a culture that was established that would allow those kinds of skills to be developed and Andy's referred to them slightly obliquely.

It was a dark culture.

It was and it has to be said it was a very homophilic culture.

There were no women allowed in this culture.

So the, you know, in my time, the culture was really one man and adoring young men.

And you had to repeatedly affirm that adoration.

And part of that was really useful and really enjoyable.

Andy's already mentioned the lunches that exactly the same regime, you know, characterized my stay.

There was a rigorous routine where all the young men would go off to the New York cafe, would have onion rings and burgers and Edelman would wax lyrical.

And perhaps this is not art and creativity, but he was he was he was artistic in his joke telling.

So he would always have a new joke every every lunchtime in a New York diner.

And that was a routine and it was just pure theatre day after day after day, incredible energy.

But it was theatre.

The young men were the audience.

And we basically, I think, were dancing to his orchestration.

He was the conductor and he was the director.

If that culture, if that is possible, that that culture did actually instill in all of us, in particularly Olaf and Julia, the notion that to make it to make a difference and to be true to one's convictions, you really have to sort of fight for and autonomously pursue what you want to do.

And it is interesting that the style in which both Julia and Olaf have pursued their scientific ideology and ideas and creativity is very much, it's slightly lonesome.

With the exception of Julia's work in the molecular biology of sleep, the theoretical contributions are very associated with them personally, to a certain extent with me.

And what one could argue is the mathematical version of value-dependent learning, which would be the free energy principle.

All of these things have the same flavor as neurodarminism or as some people used to call it, neuroedelmanism.

So I often ask that question, was this the skill of Edelman in choosing people who are going to make a difference?

Or did he basically provide a culture in which it gave people the tools or the motivation or the confidence that you can make a difference?

Because you saw him make a difference.

I don't know.

I think both of you actually said something.

I shared the experience.

Definitely, he set up a right environment.

But just like when we do same cell, same cell and its niche, you need the interaction.

You need the conversation between epithelial and mesenchymal cells.

Or in the brain, I heard it's more like interaction between neuronal cells and the Julia cells.

So a successful interaction would lead whatever the process to move forward.

But not every process would be successful.

So as I said, I also like the way Kyle mentioned about Edelman.

He's the conductor.

He actually said that in our meeting at the time.

He said, Edelman, I'm the conductor.

You know, you guys, you know, it's my symphony players.

And you pray together, you can make a good music.

And but it's also true.

Not every interaction works.

Because I was with him for almost eight years or so.

Not every person works.

Let's put it that way.

That is a fact.

So some work, some don't.

Right.

So I think people are given that opportunity.

And as Andrew said, it's a good outstanding environment.

So you pick up the idea from Edelman.

You develop yourself.

And some of them, you know, or you can call it Edelman.

Then we success in a certain way or not.

But I think it's an exciting environment.

I would put it that way.

It's both.

You know, he inspires you.

He gives you a sort of exciting environment to evolve.

But not everyone is chosen.

He can choose everyone to be successful.

No, it's not like that.

But he did give you that opportunity to be successful.

And to pick up something.

And he also gives you that courage to do it.

As I mentioned, you know, people would normally think, you know, using the feather as a model is kind of very, very high risk.

But he has that encouragement.

And I do that.

And I feel I really using feather as a Rosetta stone.

I have learned something for the morphogenesis pattern formation field.

Yeah.

So I appreciate him for that.

Yeah.

Andrew, or I can ask another question.

I really don't have anything to add right now.

Thank you, though.

It's really amazing.

It's such a different world.

And the environment of learning and of carrying out science is so different.

So personally, I really appreciate just even a window into settings that are elsewhere and in a different time.

So thank you for all of this.

All right.

I'll ask another question from Dave.

Professor Edelman was reputed to be skeptical about the risk of abusing mathematical or computer models in creating theories of neural functioning, cognition and consciousness.

Some reviewers went so far as to mock him in print for using computers as research tools.

But he worked with Tonini on the foundations of integrated information theory, an extremely math heavy theory that sometimes explained using computing imagery.

Did Edelman change his attitude towards the role of math or of computing paradigms and thinking about the mind?

Or did he simply start trusting his colleagues to think computationally without falling into rabbit holes dug by von Neumann and Turing?

Or how did you see the role of math and computers in Edelman's thinking and research develop over a period where they were also undergoing such rapid developments more broadly?

So I'm pretty sure that Edelman respected mathematics and computational approaches to neuroscience.

He certainly there certainly were a lot of theory fellows that were computationally savvy while I was at the Institute.

And so, for instance, Carl mentioned these robots that were used to sort of explore and learn and sort of the beginning, I guess, of embodied cognition, Carl, you would say, where they explore the environment and learn.

And then from that learning, they build internal models.

So that is, I would say, and that was Olaf's Ballywig.

I would say that that was highly computational and certainly a major thrust in Edelman's emphasis and what he wanted to do at the Institute.

And there were a number of other things.

I think the integrated part that Giulio did, I think that was almost all Giulio's work along with some of the other computational fellows.

So, you know, I think Edelman wanted to be part of that, but that really, I think, was mostly Giulio's.

And I think, you know, as part of the culture, Edelman managed to get his name on Giulio's paper.

We used to laugh about that quite a bit.

But, you know, again, I would say that was 99% Giulio's work.

Yeah, I can compare because I was there when the work was actually done.

There was a sort of period, I can't remember, there was some hypothesis that pertained to a core, a dynamical core, as sort of a prescience, unnecessary for some kind of consciousness.

And I think Edelman was comfortable with this sort of intuitive notion of a dynamic core.

But when you when it became articulated in terms of information theory, he started, he would, I think, even if he didn't say so out loud, he would have responded as he responded to me with, oh, you've got mathematosis.

So this is this is another of his ways of dealing with people who had a tendency to over mathematicalize and oversimplify as a physicist toward, you know, reducing a horse to a point and then writing down differential equations.

He was very allergic to that.

You know, he didn't like that and would call that an instance of mathematosis.

I think that's very distinct from what Andy was just talking about, which is the commitment to physically realized embodied computation.

And, you know, I think it's a really well observed point that this was embodied cognition circa 1980, 1990.

So he was quite visionary in, I think, sort of testing out ideas in physical and realized systems.

You couldn't just hand away with maths.

You had to show it worked.

This idea worked in a real in a real setting.

And at that time, they were, you know, the Darwin series.

And, you know, to the extent that, you know, I, for example, had to spend six months learning to paralyze C in order to program the robot.

So, you know, it wasn't he was frightened of new technology or computer science.

I think he didn't like trivializing things, you know, a la physicist with mathematics.

But you would certainly be fully committed to the notion that computation is at the heart of the kind of dynamics that he wanted to understand and underwrote theory.

And indeed, I think that, you know, the theory of neural group selection speaks that he wanted to see it realized, you know, realized in a robot.

So, but I think Andy's absolutely right.

At the point when Julia sort of just basically went straight from the axioms to consciousness and phi with equations.

I think that was the time that Julia had to had to leave the family, basically.

And Julia did so.

So to extrapolate that a little bit more, I think, you know, when I read Edelman's books and some of his papers, I don't understand his words.

OK.

And, you know, his phrases and his descriptions, you know, a lot of things don't make sense.

If he had written those out or someone had written those out as equations.

OK.

You know, that's the nice thing about equations.

Once you see the equation, you know, you don't have to worry about the word so much.

You know, things become pretty clear.

And I think, you know, looking back, if he'd been able to do that, I think his ideas, first of all, would have been crystallized in his own mind better.

And second of all, would have been able to be others would have understood them more clearly.

And I think that's really important.

If you have for me, OK, maybe with an engineering background or something, you know, if you have a model or if you have an idea, if you can write that out explicitly with an equation, it really clarifies things.

It crystallizes things.

And that was the thing that I liked about Julio's work, because, again, this word of consciousness, you know, it's like nobody knows what that really means.

But at least Julio had a form of equations, right, where he was talking about information theory and the way, you know, you could have a cluster where information was internal versus the external information.

And say that, OK, if you have to have consciousness, here it is.

And you could describe that with a concrete equation.

I mean, that's a step forward.

And again, I always say as someone that that records neurons in the brain all the time, if I wanted to find consciousness, how would I know it if I found it?

Right.

I mean, if I had an equation, maybe I could do it.

But, you know, just the way it's defined right now.

I mean, who knows?

I could have already discovered it.

I wouldn't know.

So I think that's an interesting point anyway.

I can't resist his following up on that, because especially evident for me in the remembered present, this notion that his words don't actually make any sense.

I think that's pure poetry and pure brilliance, because what I tend to find myself doing is reading the meaning into his ambiguous sentences and finding all sorts of meanings, which wouldn't have occurred to me otherwise.

I think that's part of his art.

I suspect that he would have been terrified if somebody had actually come along and tried to time down to a clear, formal, naturalised mathematical formalism, because he wouldn't have had that sort of biotic expressiveness and ambiguity that, of course, is at the heart of his evolutionary thinking.

I think it's great that you're able to say you don't understand what he wrote, because I look at some of his sentences.

What on earth is he trying to say here?

And of course, the exercise of trying to make sense of Edelman itself brings to the table some really interesting sort of associations and ideas.

I'm wondering whether that was part of the secret of his success, is that he resisted the temptation to be crystal clear and just left that mystical, magical ambiguity hanging in the air for people like you and me to worry about and try to resolve.

I suppose he's given that privilege after winning a Nobel Prize.

I think some of us mere mortals wouldn't be afforded that luxury.

All right.

I'll read a few more questions.

We can have a few more things.

So small questions.

In the book Topobiology, Chengming, did you draw the feathers or is that your work?

The feather is my work.

Yeah.

It's based on my work.

Yeah.

Awesome.

Well, how do we take things forward in science and in our work here in this level of the fractal with this amazing story and history with Edelman?

But how do we take our historical and our mentoring context?

Like, how do you carry that forward?

Is it something you think about?

I personally do.

And I congratulate myself every day on not being like Edelman when it comes to mentoring.

How so?

I'm not going to answer the how so question, but you can read between the lines.

Whether that's the right thing to do or not, I don't know.

I don't think you could be Edelman now.

You know, in the 21st century, we'd be too close to all the culture wars, I think, to even entertain that style anymore.

And I suspect that's quite a good thing.

Although, you know, pressure makes diamonds.

And as Andy says, it was an exciting, both my colleagues have said, it was an exciting, informative time.

It was a pressured time.

It was a dark time for me, you know, in terms of the dynamics, the systemic, the personal interactions.

But because it was pressured, it made diamonds.

And so we may be throwing away the opportunity to make academic diamonds and be truly creative in terms of making things that endure like diamonds by not having that style of mentorship.

But I don't think it would be even legal nowadays, to be quite honest, to run a lab like Edelman run.

So, again, I mean, I think that over the years, neuroscience, I don't know about the rest of the science, but it's almost become sort of industrialized.

So there's a method, you know, you as a young person, you when you finally get your own job, you have to set out.

And if you're at a university environment, there's certain boxes you have to check off.

You have to get a grant.

You know, you have to get a certain number of papers.

You have to get something in science or nature in order to get promoted.

You know, there's these steps.

And I think young people are so focused on moving through those steps at the point in their career where they're going to be the most creative and their minds are the most flexible and they can be the most productive.

And, you know, I see that as a shame, actually, that they're not given this opportunity to be a little freer in the way they accomplish things.

And then, you know, when they do accomplish a modicum of success, they rely on their past behavior.

So instead of saying, OK, I had to go through that in order to get tenure.

Now that I have tenure, I'm free and I can do something that I really want to do.

But that really doesn't happen because people get stuck in a rut.

And I see that as really holding back progress in neuroscience.

And that was kind of the cool thing about the Neuroscience Institute.

For better or worse, we weren't allowed to go out and get our own grants.

We had to rely on internal funds.

But as a result, the only person we had to convince was Edelman in order to get funded.

And to a certain extent, that allowed us to do things that were really creative.

In hindsight, I think that was a mistake because financially that was a doom for the Institute.

There could have been a hybrid approach that would have been more productive.

But then again, as Carl was saying, the investigators would have had to have more freedom to do that.

And that was not exactly in the culture.

Although I think I was an outlier because I maintained a separate lab.

So I was somewhat inoculated from that.

And that made it a lot more pleasant for me.

But again, I see that in current neuroscience, it would be nice if people were willing to take more chances and be a little more creative instead of hoeing to the status quo.

That would be, I think, what we need in order to move forward.

Yeah.

Yeah.

No, I think that in terms of the mentorship that Carl was talking about, I mean, certainly I have seen people do very well.

But I also have the opportunity, you know, sort of when I moved to Southern California, University of Southern California, to have some interaction with Dr. Seymour Benzer.

And some of you probably know him.

I mean, you would see, actually, he is also very successful.

But when he came out, he would always have his students surrounding him, you know, very, very kindly.

I think that, you know, Edmund is brilliant, but it's kind of the sun is too bright.

Sometimes people have to be distilled into the, as you say, academic diamond.

That's interesting.

So we also have seen some people, you know, didn't get along with him well.

But still, I think there are many, many impacts.

He certainly have encouraged many people to go different directions.

And in terms of the academic future, right?

I mean, I think you two are more talking about the neurobiology side.

But on the morphogenesis side, I would also say he always emphasized the patterns, you know, thinking that, you know, pattern is kind of the way nature, you know, either biology or non-biology.

They leave some clue for us to figure out.

And when I was there, I remember every time he talked about the visual columns, because we saw they have worked out.

He got very excited.

And in fact, I have one of my peers at the last time was working on the visual columns of the cat.

But I think that paper, that work never really go out to be published.

But anyway, on a more practical thing, I work on the tissue patterns.

And I feel that, you know, you mentioned about the topobiology.

And again, I think he coined the term.

And I think the term, I mean, I think it means a lot because I kind of, you know, eventually my career was trying to figure out these signaling molecules.

Because at that time, people only talk about the signaling molecules, very, very important.

But the idea of topobiology is that you even position these simply by positioning these signaling centers in a different topology.

You can actually have a lot of consequences.

So from there, the signaling center can also have a different lifetime, the strength and the duration and the range they can influence.

So by modulating, you know, he likes these terms too, modulating these signaling centers, you can create a lot of regional differences.

And up to now, as I say, from my own take, you know, bird, you look at it, fried feather here, downy feather here, you know, the beautiful crown feather here, tail feather.

But these are all from the same genome.

So it actually is through the positioning of these different signaling centers in a different way.

And you basically have the same ingredient, but you can play with them by topologically temporal and especially arrange them and to create a quite complex pattern.

So, so, so I, I feel that in the tissue pattern in morphogenesis side, you know, I, I feel that, you know, his vision is, is great.

And I, I think we still try to work out more about how these patterns are positioned.

But I think that's sort of the direction in a more visualized topobiology aspect, we can appreciate.

Thank you.

Awesome.

Well, perhaps if each of you would like to give any closing thoughts.

Yes, I, I'll, I'll just start, I'll just do it to echo your, those last sentiments.

I think everything that Edelman said or tried to say, depending whether you could understand him, I think he's absolutely right.

And we are slowly discovering the truths.

We just heard about topobiology and the importance of patterns.

And of course, if you speak to somebody like Mike Levin, he'd be talking about exactly the same thing.

He'd use words like basal cognition and, you know, pattern formation.

But I think that we're talking about the same, the same principles of biological self-organization.

You know, and Edelman, I didn't know about topobiology, but I do know, you know, there are other fundamental ideas that he just, you know, incidental to him.

I am absolutely sure.

I think notions of degeneracy in the neurosciences and psychology.

You know, the notion of many to one structure function mappings, for example, these are fundamental ideas which now are taken almost for granted in, in, in my world.

But, you know, you sort of forget how much he actually contributed or at least laid the seeds for and sometimes very explicitly.

So it should be acknowledged, although he was, as Mark Rakel put it in his typically gentlemanly fashion, Edelman was a very complicated man.

He was also very brilliant and is responsible, I think, for a lot of the direction of travel and the intellectual growth, certainly of my intellectual growth, but I would say also many aspects of, of, of our academic communities.

Thank you, Carl.

Perhaps Andrew.

Oh, wait, unmute and then please continue.

Yeah.

So I would say, you know, if there's anyone listening out there, there's something to be said about maybe looking back in the ancient literature that goes back quite a bit and maybe seeing how some of these things came about and trying to put it together.

And see that, you know, there's this iteration that constantly takes place.

People rediscover the same thing that's been discovered multiple times.

And, you know, try to see how there is a status quo in the way science works.

And then there's these outliers where there's sort of creative bubbles that come to fore.

And I think it's interesting to think of Edelman and the things he did and the people that were associated with him in terms of sort of these disruptive, right?

They call it disruptions.

And how those disruptions may have consequences many years later.

It's kind of interesting to take the long view.

So I think it's maybe worthwhile thinking about this once in a while.

And for young people out there to be encouraged to think off to think out of the normal way of doing things once in a while.

Try to foster your creativity and have the courage to to go after some of those ideas.

I think that's a take home message for me anyway.

Yeah.

I would say I always like, you know, all sort of, you know, influenced by his idea saying that we should study the important question that have a general meaning.

So I think that kind of idea, he said, look at the big picture.

I remember when I was there, he talked like that, big pictures.

So that is where when he thinking about the Darwinism, he would say it's not only for the species evolution, for the immunoglobulin coronal selection.

He would push it to the brain theory.

And for me, I work on the morphogenesis.

I actually would come to realize that is also what happened at the tissue level.

These cells are competing with each other to form that pattern.

And then pattern is selected when there is a functional advantage.

It will be stabilized.

So I think the idea of saying if we're going to do science, we ask big questions, ask for the generalization.

I appreciate that.

And certainly he is like how you want to call it, right?

A huge elephant.

So we all touch him through different sides and have a different take on that.

And in terms of the young scientist, I also remember he likes the Chinese proverb.

I cited to him, that is to say, if you want to get a tiger's son, you must get into tiger's cave.

So we three did.

Thank you all.

That's a great note.

Really appreciate this conversation.

And again, thanks until next time.

Okay.

Thank you.

Bye.

Nice meeting you all.

Bye-bye.

Thank you.

Bye-bye.